

FINAL INTERNSHIP REPORT

Internship Report

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Program: : Artificial Intelligence and Data Science

Company Name: : Astana IT University

Department: Artificial Intelligence and Data Science

Position: Scientific Research Intern

Workplace Advisor: Beibit Abdikenov

Advisor Position: Director of Science and Innovation Center "AI"

Internship Duration: : March 9, 2026 - May 9, 2026

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1. Introduction

This report presents the work completed during my internship in the field of data analysis and machine learning. The primary objective was to develop a machine learning pipeline for classifying breast cancer histopathology images using the BreakHis dataset.

The internship focused on applying concepts from my academic program, including data preprocessing, exploratory data analysis, and deep learning model development.

2. Workplace Description (MANDATORY)

Company Name: Astana IT University

Location: Astana, Kazakhstan

Industry: Information Technology / Data Science

The company operates in the field of data analytics and software development, providing solutions related to data processing, machine learning, and system optimization.

Department:

Data Science / Machine Learning Department

Department Role:

- Develop predictive models
- Analyze large datasets
- Build data pipelines
- Support decision-making using data

Technologies Used:

- Python
- PyTorch
- Pandas, NumPy
- OpenCV
- Jupyter Notebook / Google Colab

My Role:

Data Science Intern

Responsibilities:

- Data preprocessing
- Feature extraction

- Model development
- Performance evaluation
- Documentation

3. Internship Objectives

The main objectives of this internship were:

- Understand real-world data processing workflows
- Build a machine learning model for medical image classification
- Apply data analysis and deep learning techniques
- Evaluate model performance using appropriate metrics

4. Technical Work and Tasks

During the internship, I worked on the BreakHis dataset.

Tasks performed:

1. Data Loading
2. Metadata Extraction
3. Exploratory Data Analysis (EDA)
4. Data Preprocessing
5. Model Development (CNN)
6. Model Training
7. Model Evaluation
8. Error Analysis

5. System Architecture & Implementation

◆ Pipeline Overview:

Raw Images → Metadata Extraction → EDA → Preprocessing → CNN Model → Evaluation

Overall Machine Learning Pipeline



Figure 1: Overall machine learning pipeline used for breast cancer histopathology image classification.

Model Architecture

The model used is a Convolutional Neural Network (CNN):

- Input: 128×128 RGB images
- Conv Layer 1 → ReLU → MaxPool
- Conv Layer 2 → ReLU → MaxPool
- Fully Connected Layers
- Output: 2 classes

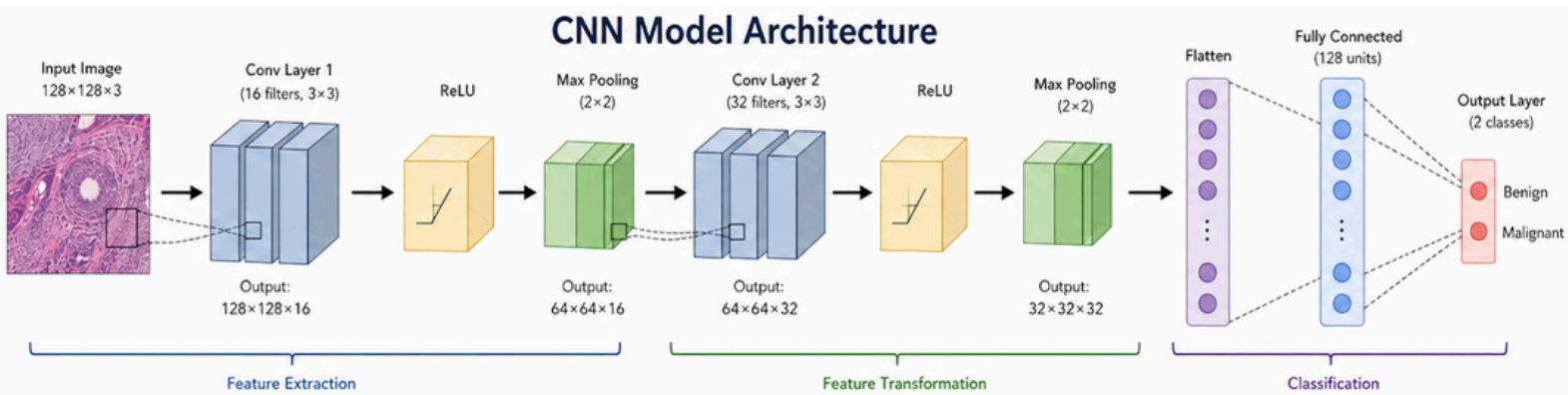


Figure 2: CNN architecture used for breast cancer classification.

Code Snippet (Example)

```
class SimpleCNN(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv = nn.Sequential(
            nn.Conv2d(3, 16, 3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2),
            nn.Conv2d(16, 32, 3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2)
        )
```

6. Tools and Technologies

Tool	Purpose
Python	Programming
PyTorch	Deep Learning
Pandas	Data handling
NumPy	Numerical operations
OpenCV	Image processing
Matplotlib/Seaborn	Visualization
Google Colab	Execution environment

7. Challenges and Solutions

Challenge 1: Limited Dataset

- Only a few patients available
- Solution: Used patient-wise split and reduced dataset

Challenge 2: Slow Training

- Solution: Enabled GPU (Tesla T4)

Challenge 3: Low Accuracy

- Caused by small dataset
- Solution: Analyzed errors and explained limitations

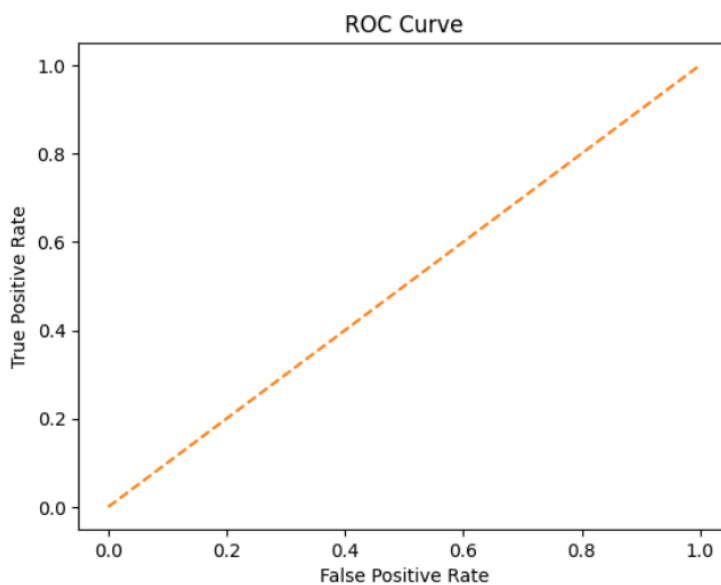
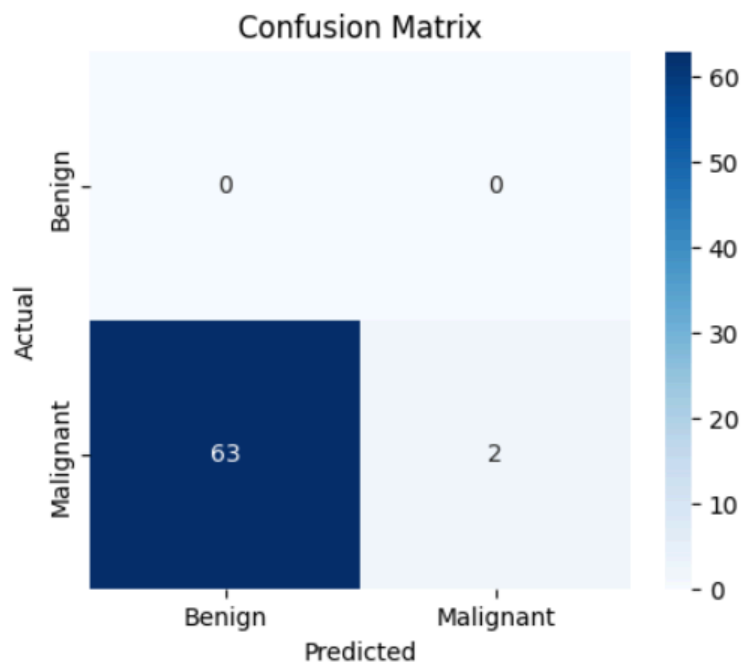
8. Results and Evaluation

Metrics used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC

Observations:

- Model performance is limited due to small dataset
- Misclassifications observed between classes



9. Evidence of Work (VERY IMPORTANT)

- Jupyter Notebook implementation
- Code for CNN model
- Visualizations (EDA plots)
- Model evaluation outputs

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
import os
import random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import cv2
import torch

random.seed(42)
np.random.seed(42)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Device:", device)
```

Device: cuda

```
dataset_path = "/content/drive/MyDrive/BreaKHis_v1/histology_slides/breast"
```

```
image_paths = []

for root, dirs, files in os.walk(dataset_path):
    for file in files:
        if file.endswith(".png"):
            image_paths.append(os.path.join(root, file))

print("Total images:", len(image_paths))
```

Total images: 7909

epochs = 2

```
for epoch in range(epochs):
    model.train()

    for i, (images, labels) in enumerate(train_loader):
        images, labels = images.to(device), labels.to(device)

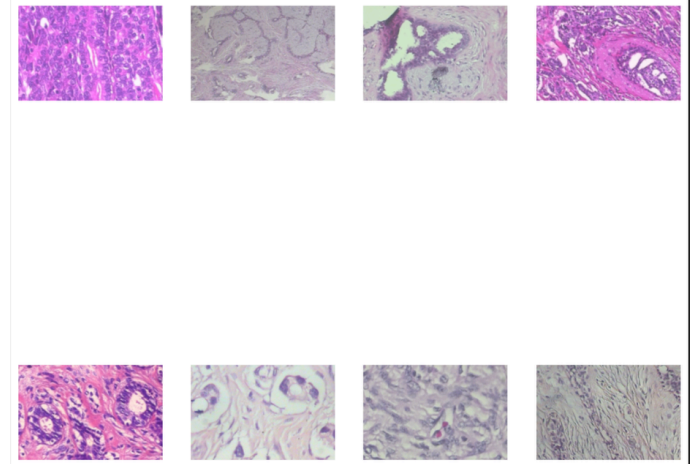
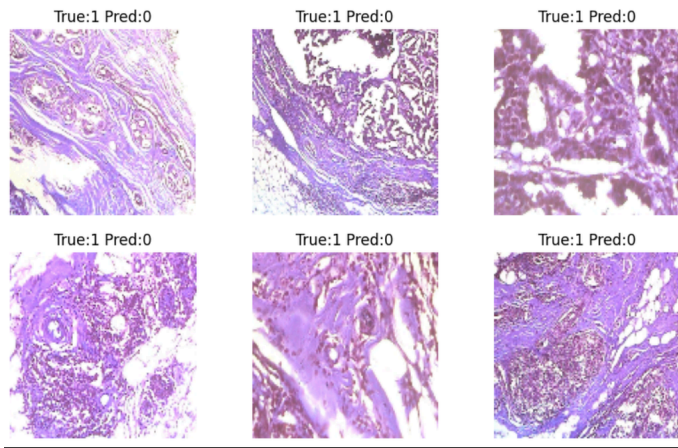
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)

        loss.backward()
        optimizer.step()

        if i % 50 == 0:
            print(f"Epoch {epoch+1}, Step {i}, Loss: {loss.item():.4f}")

    print(f"Epoch {epoch+1} DONE")
```

```
Epoch 1, Step 0, Loss: 0.7027
Epoch 1, Step 50, Loss: 0.4600
Epoch 1 DONE
Epoch 2, Step 0, Loss: 0.4855
Epoch 2, Step 50, Loss: 0.3047
Epoch 2 DONE
```



10. Reflection

During this internship, I gained practical experience in applying machine learning techniques to real-world datasets.

Key Learnings:

- Importance of data preprocessing
- Handling real-world data limitations
- Understanding model evaluation metrics
- Working with deep learning frameworks

Challenges:

- Limited dataset size
- Computational constraints

Growth:

This internship improved my understanding of how machine learning systems are developed and evaluated in real-world scenarios.

11. Conclusion

This internship successfully demonstrated the application of machine learning techniques for medical image classification.

Although the model performance was limited due to dataset constraints, the project provided valuable insights into data analysis, model building, and evaluation.

Future improvements include:

- Using larger datasets

- Applying advanced models (ResNet)
- Improving generalization

12. References

- BreakHis Dataset (UFPR)
- PyTorch Documentation
- Scikit-learn Documentation
- Research papers on medical image classification